

ICST INETRDISCIPLINARY CENTER FOR SMART TECHNOLOGIES

Center of CS Department's Software Engineering Track

SMART PARKING
ESP32 · IoT

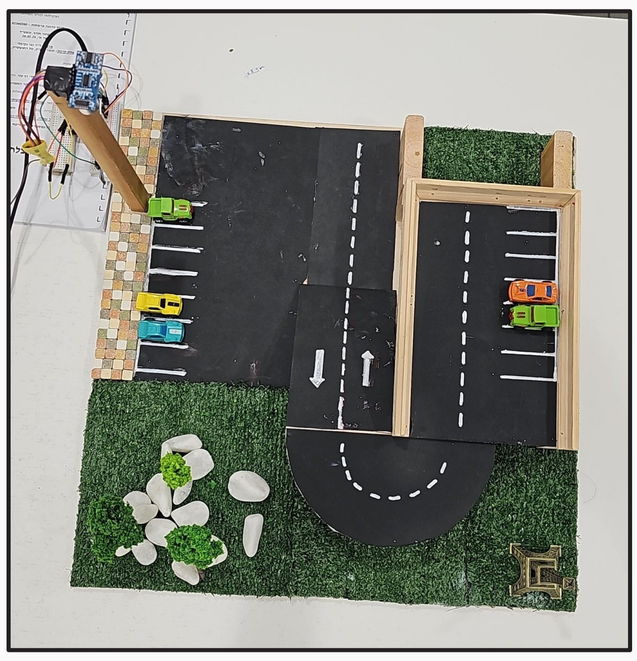
Smart Parking

The nearest free spot, every time

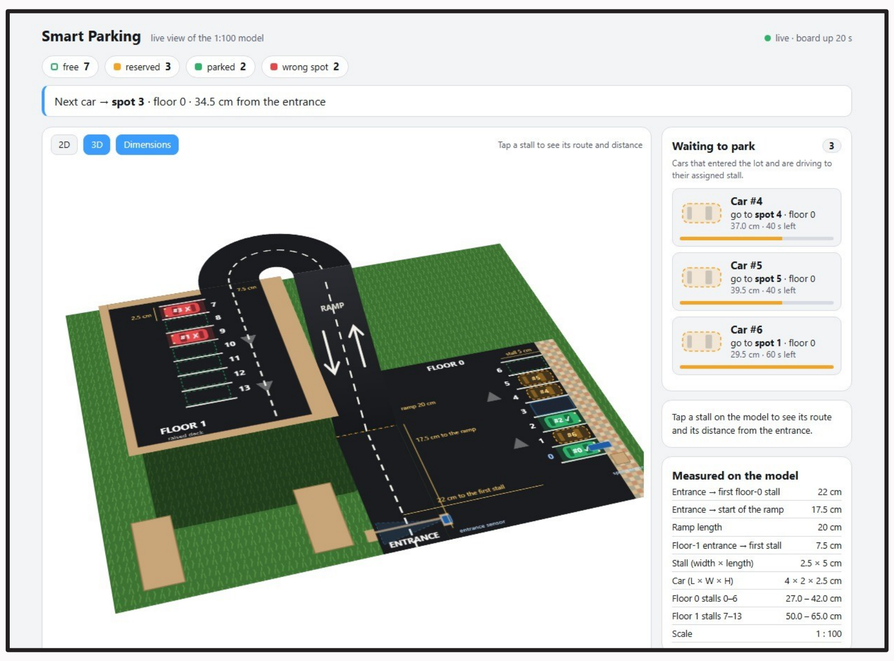
Drivers in large parking lots waste time and fuel searching for a free spot.

Smart Parking is a two-floor, 14-spot lot run by an ESP32 that sends every arriving car to the nearest free spot:

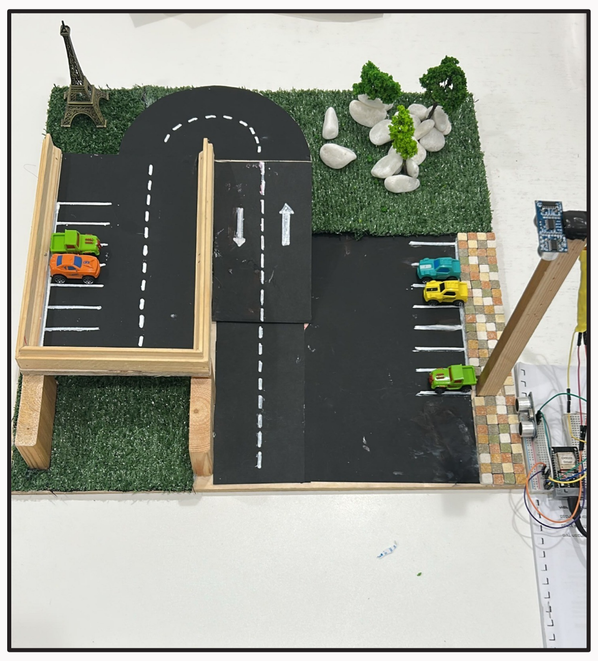
- **Nearest-spot assignment** using A* path search
- **Ultrasonic sensing** of arriving and parked cars
- **LED alert:** steady = right spot, blinking = wrong
- **Live web map** served by the board – no internet



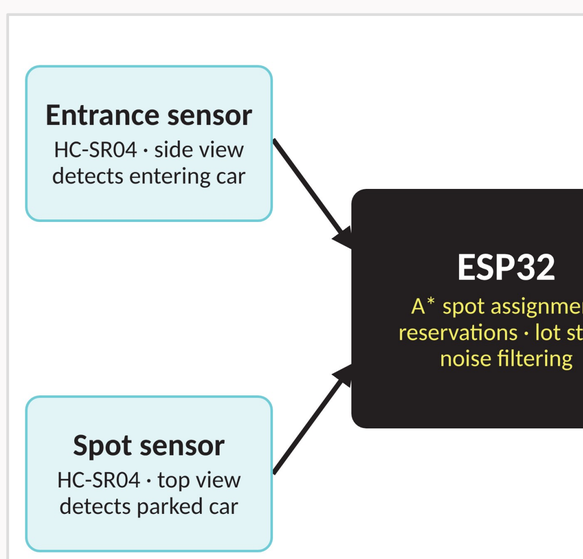
Prototype – top view



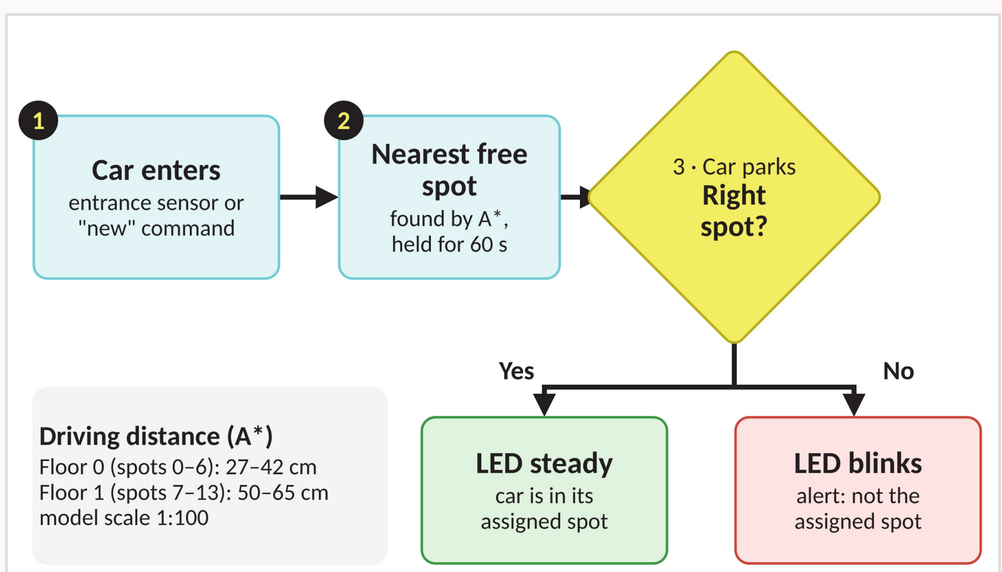
Live web view – 3D map



Prototype – front view



System diagram – sensors, ESP32 and outputs



How a car is handled – from entry to LED alert

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Technion – Faculty of Computer Science

A Project in Internet of Things (IoT)

Computer Science Department

הטכניון
מכון טכנולוגי לישראל